Project 2026

Proton Transfer in (mixed) Dimers and Trimers of HCl and H_2O

In this project, we will examine the structures, interactions and multiple proton transfer reactions in dimers and trimers of HCl and H_2O, as well as in the mixed dimer $(HCl)(H_2O)$ and in the mixed trimers $(HCl)_2(H_2O)$ and $(HCl)(H_2O)_2$. Structures will be computed using density functional theory, followed by high-level *ab initio* energy calculations. The systems will be studied either in the gas phase or in an implicit solvent model. The density functional and "gas phase" or "implicit solvent" will be assigned.

1. Compute the structures with Gaussian 16, using the assigned functional and the **Def2SVP** basis set. Apply symmetry where appropriate. Use **Opt Freq=NoRaman** for the minima and **Opt=(TS,CalcFC) Freq=NoRaman** for the transition structures. For implicit solvent calculations, add the keyword **SCRF=PCM**. Water is the default solvent.
   Check the number of imaginary frequencies. The checkpoint files will be used in the following steps.
2. Refine the structures using the **Def2TZVP** basis set, using the checkpoint file, and specify **Guess=Read Geom=Check** as usual. Use **Opt=ReadFC Freq=NoRaman** for minima and **Opt=(TS,ReadFC) Freq=NoRaman** for transition structures and possibly for structures with higher symmetry (those with NImag=1 in step 1). For implicit solvent calculations, add the keyword **SCRF=PCM**. Specify **Pop=NBORead** and the $NBO keyword line to obtain the gas phase NPA charges and NLMO bond orders.
3. Perform *gas phase* counterpoise calculations for the lowest minima (dimers and trimers) from step 2. The implicit solvent model cannot be used in counterpoise calculations.
4. Calculate high-level energies (single points) with Orca at the **DLPNO-CCSD(T) aug-cc-pVTZ** level using the structures computed in step 2. Activate the **NoFrozenCore** option. For implicit solvent calculations, add the keyword **CPCM(Water)**.

Starting points for transition structures can easily be constructed from the corresponding minima by moving the hydrogen atoms to obtain a symmetrical structure (apply symmetry); no interpolation is required.

The structures of $(HCl)_2(H_2O)$ and $(HCl)(H_2O)_2$ differ significantly between gas phase and implicit solvent (see the structures below). In implicit solvent, an optimization may halt due to a linear arrangement of atoms; check the output. Should this happen, then restart the calculation from the last coordinates using **Opt=Cartesian** without **ReadFC**; this is slower but likely to alleviate this problem.

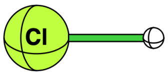
Enthalpy corrections, based on statistical thermodynamics, are found in the **jsum** output of a frequency calculation under `Thermal corrections: Enthalpy`.

How to write the report:

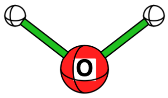
- write a concise report, don't write long texts to hide your ignorance 😉
- Introduction, methods, results, discussion, conclusions, as usual
- Include tables:
  - "raw" data: DFT total energy (a.u.), NImag and enthalpy correction (kcal/mol), DLPNO-CCSD(T)
  - relative energies and relative enthalpies (kcal/mol or kJ/mol)
- Evaluate the interaction enthalpies within the dimers and trimers **3**, **5**, **7**, **9**, **11**, **13** and **15**
- How serious is BSSE for these systems?
- Compute the activation energies for proton transfer
- Provide pictures of the computed structures. Pro-tip: use "constrain proportions" when scaling.
- Include relevant distances in the pictures (in Molecule: double-click a bond or add a "line" bond).
- Optional: compare structures, charge distribution, interaction energies and activation barriers in the gas phase and in solution.
- Give literature references where appropriate.

Structures

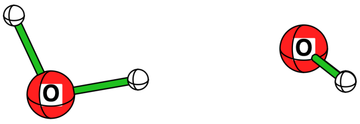
1 HCl



2 H_2O

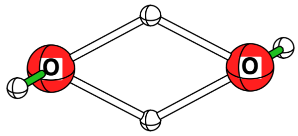


H_2O dimer



3a $(H_2O)_2$ C_s

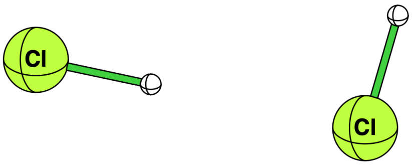
3b $(H_2O)_2$ C_i



4 TS $(H_2O)_2$ C_{2h}

Structure **3a** may be a TS with Def2SVP and **3b** the minimum. This reverses with Def2TZVP.

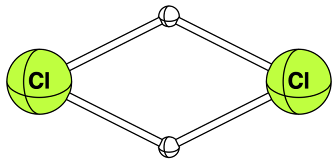
HCl dimer



5a $(HCl)_2$ C_s

5b $(HCl)_2$ C_{2h}

5c $(HCl)_2$ $C_{\infty h}$

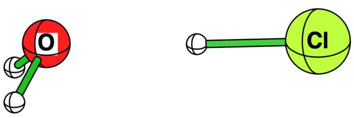


6 TS $(HCl)_2$ D_{2h}

5a may not be found with Def2SVP, it is the minimum with Def2TZVP.

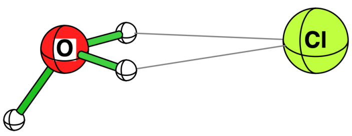
The linear structure **5c** is NImag=2, use **Opt** without **ReadFC** in the refinements.

H_2O HCl mixed dimer



7a H_2O HCl C_s

7b H_2O HCl C_{2v}

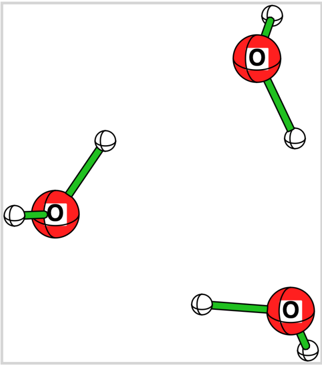


8 TS $H_3O^+Cl^-$ C_s

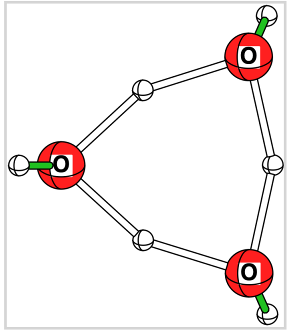
7a and **7b** are close in energy, **7b** may be the minimum with Def2SVP

H_2O and HCl trimers

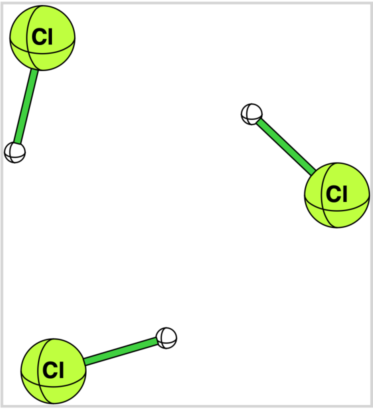
9 $(H_2O)_3$ C_1



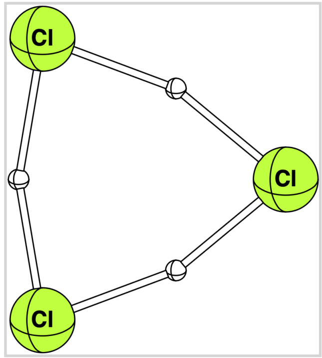
10 TS $(H_2O)_3$ C_s



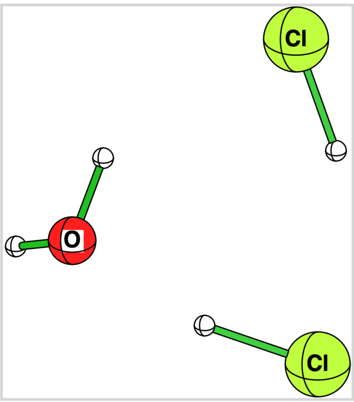
11 $(HCl)_3$ C_{3h}



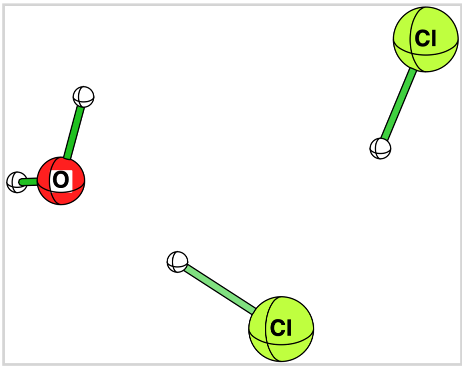
12 TS $(HCl)_3$ D_{3h}



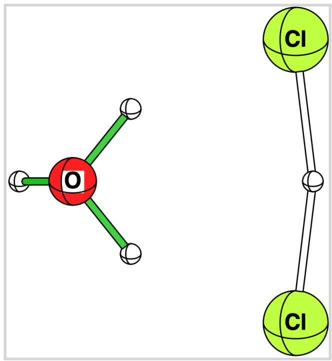
Mixed trimers



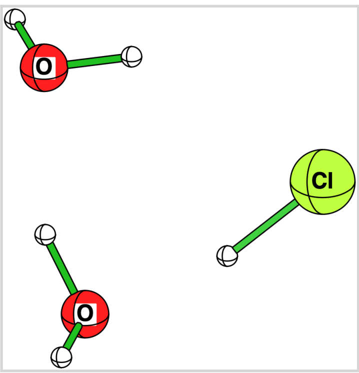
13 $H_2O(HCl)_2$ C_1



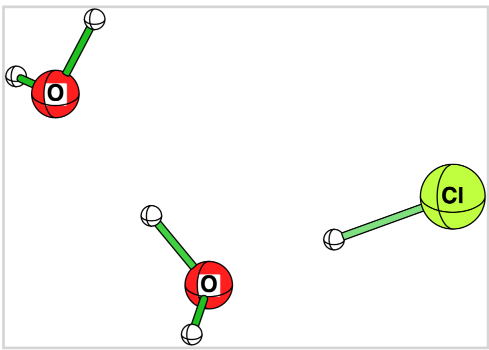
13 $H_2O(HCl)_2$ in implicit solvent



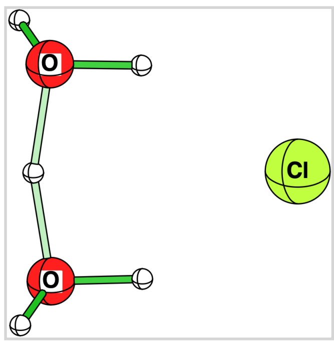
14 TS $(H_3O)^+(HCl_2)^-$ C_s



15 $(H_2O)_2HCl$ C_1



15 $(H_2O)_2HCl$ in implicit solvent



16 TS $(H_5O_2)^+Cl^-$ C_2